Assignment - 4

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Problem Statement : Write a C program to store first year % of student in an array. Write a func. for sorting array of floating point no. in ascending order using Bucket sort & display top five scores.

Objective :
1) Understand Bucket sort
2) Implement Bucket sort using func.
3) Sort floating point percentage
4) Display top five scores
5) Analyze complexity

Implementation :

Platform — 64-bit open source linux.

Test Condition —

Test case :	Input	Expected
TC1	One percentage	Stored successfu
TC2	Multiple percentage	Stored correctly

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TC3	Random floating values	Ascending o
TC4	Duplicate values	Duplicate re
TC5	< 5 Students	Display available
TC6	>= 5 Students	Display top 5 s

Pseudo Code :

1) START
2) Read no. of student (n)
3) Create n empty buckets
4) Read % of each student
5) For each %
   (a) Find the appropriate bucket
   (b) Insert % into that bucket
6) Sort such each bucket individually
7) Merge all bucket into a single sorted array
8) Display the sorted %
9) Display the top five highest %
10) STOP

Time Complexity :

- Accepting Input : O(n)
- Creating & Filling bucket : O(n)
- Sorting individual buckets : O(n)
- Merging Bucket : O(n)
- Displaying Top Five : O(n)

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Overall average Time complexity : O(n)

Conclusion : Bucket sort efficiently sorts the distributed floating point percentage & helps identify the top five performs.

FAQs :

1) What is bucket sort ?
Bucket sort is a sorting algo. that divides elements into diff. buckets, sort each bucket individually & then merge them to produce the final sorted list.

2) Why use it for floating point data ?
It works efficiently for floating point no. because value are usually uniformly distributed over a range, allowing them to be placed into appropriate buckets for faster sorting.

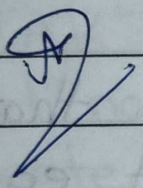
3) What is a bucket ?
A bucket is a temporary container that stores elements belonging to a specific range of values before they are stored.

4) Worst case complexity ?
$O(n^2)$, when all elements are placed into the same bucket.

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5) Application ?

- Sorting percentage or marks.
- Sorting floating points no.
- Data analysis & statistics.
- Large database with uniformly distributed values.



```c
#include <stdio.h>

void bucketSort(float arr[], int n)
{
    int i, j;
    float temp;

    for (i = 0; i < n - 1; i++)
    {
        for (j = 0; j < n - i - 1; j++)
        {
            if (arr[j] > arr[j + 1])
            {
                temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }
}

int main()
{
    int n, i;
    float arr[100];

    printf("Enter number of students: ");
    scanf("%d", &n);

    if (n <= 0 || n > 100)
    {
        printf("Invalid number of students.\n");
        return 1;
```

```
        return 1;
    }

    printf("Enter student percentages:\n");
    for (i = 0; i < n; i++)
    {
        scanf("%f", &arr[i]);
    }

    bucketSort(arr, n);

    printf("\nPercentages in ascending order:\n");
    for (i = 0; i < n; i++)
    {
        printf("%.2f ", arr[i]);
    }

    printf("\n\nTop five scores:\n");
    for (i = n - 1; i >= (n > 5 ? n - 5 : 0); i--)
    {
        printf("%.2f ", arr[i]);
    }

    printf("\n");
    return 0;
}
```